30 Supergraphs

30.3 Calculations with Supergraphs

We now consider how to use the results of the previous sections to calculate the quantum effective action $\Gamma[S, S^*]$ for a set of classical potential superfields $S_n(x, \theta)$ and their adjoints. Following the prescription discussed in Section 16.1, this may be defined in terms of a sum over all connected one-particle-irreducible supergraphs, consisting of vertices to which are attached directed internal and external lines. For each external line of type n ending or beginning in a vertex labelled x, θ, we include a c-number factor $S_n(x, \theta)$ or $S_n^*(x, \theta)$, respectively (but no propagator). A vertex labelled x, θ with N incoming or N outgoing lines labelled $n_1, n_2, \ldots, n_N$ yields a factor equal to i times the coefficient of the $S_1 S_2 \cdots S_N$ term in the superpotential $\widetilde{f}(S)$ or times the complex conjugate of this coefficient, respectively. An internal line of any type coming out of a vertex labelled x, θ and going into a vertex labelled x', θ' yields a propagator factor, given by Eqs. (30.2.10) and (30.2.17) as

$$-\frac{i}{4}\delta^4(\theta - \theta')\Delta_F(x - x') \,. \tag{30.3.1}$$

In addition, as shown by Eq. (30.1.7), a superderivative $\mathcal{D}_R^2$ acts on the propagators or external line S-factors of all but one of the internal or external lines coming into any vertex, and a superderivative $\mathcal{D}_L^2$ acts on the propagators or external-line S^*-factors of all but one of the lines coming out of any vertex. The product of these factors must be integrated over all xs and θs; the quantum effective action is the sum of these integrals for all one-particle-irreducible graphs.

By integrating by parts in superspace, the $\mathcal{D}_L^2$ and/or $\mathcal{D}_R^2$ operators accompanying any one propagator (say, one that connects vertices labelled x, θ and x', θ') may be moved to other propagators or external line factors. This leaves the factor contributed by this internal line proportional to a factor $\delta^4(\theta - \theta')$. Integrating over θ' then eliminates this delta function, and lets us replace θ' with θ everywhere else. (In cases where several internal lines connect the same pair of vertices, we need to use the property of fermionic delta functions that $[\delta^4(\theta - \theta')]^2 = 0$.) Continuing in this way, we wind up with a single four-dimensional θ integral, with all $\mathcal{D}$s acting on the external line factors S_n and S_n^*. That is, though not usually local in spatial coordinates, $\Gamma[S, S^*]$ is *local in the fermionic coordinates.*

The structure of this functional is governed by its invariance under the 'gauge' transformations (30.2.4). This tells us that every one of the external line factors S_n or S_n^* must be acted on by an operator $\mathcal{D}_R^2$ or $\mathcal{D}_L^2$, respectively, with two possible exceptions. The exceptions are that a term with only incoming or only outgoing external lines, in which all but one of the external line factors S_n or S_n^* are acted on by $\mathcal{D}_R^2$ or $\mathcal{D}_L^2$, respectively, and by no other superderivatives, is still invariant under the transformation (30.2.4), even though it cannot be expressed as a four-dimensional θ-integral of a functional only of the Φ_n and/or Φ_n^* alone. This is because the change in such an amplitude under this transformation could only come from the change in the external line factor S_n or S_n^* that is not acted on by an operator $\mathcal{D}_R^2$ or $\mathcal{D}_L^2$, and this change is eliminated if we use integration by parts to move one of the other $\mathcal{D}_R^2$ or $\mathcal{D}_L^2$ operators to act on it. As we saw in Section 30.1, such a term in

$\Gamma[S,S^*]$ is the $\mathcal{F}$-term of a functional of the Φ_n or of the Φ_n^*, and thus makes a correction to the superpotential or to its complex conjugate. Aside from these exceptions, every term in $\Gamma[S,S^*]$ may be written as a four-dimensional θ-integral of a functional only of the Φ_n and/or Φ_n^*, and therefore makes a correction to the D-term part of the effective action.

Also note that a term in the quantum effective action in which all but one of the external line factors S_n or S_n^* is acted on by $\mathcal{D}_R^2$ or $\mathcal{D}_L^2$, respectively, and also by additional $\mathcal{D}_R^2$ or $\mathcal{D}_L^2$ operators (such as in combinations like $\mathcal{D}_R^2\mathcal{D}_L^2\mathcal{D}_R^2 S_n$ or $\mathcal{D}_L^2\mathcal{D}_R^2\mathcal{D}_L^2 S_n^*$) may also be written by integration by parts as a term in which one of the extra $\mathcal{D}_R^2$ or $\mathcal{D}_L^2$ acts on the previously undifferentiated external line S_n or S_n^* factor. Such a term may therefore be expressed as a four-dimensional θ-integral of a functional only of the Φ_n and/or Φ_n^*, and therefore just makes another correction to the D-term part of the action. The only terms in $\Gamma[S,S^*]$ that cannot be written in this way are those that have E incoming external lines and only $E-1$ operators $\mathcal{D}_R^2$ operating on the S_n external line factors, or E^* incoming external lines and only E^*-1 operators $\mathcal{D}_L^2$ operating on the S_n^* external line factors. We can therefore tell whether a supergraph can make a contribution to the superpotential or its adjoint by simply counting the number of $\mathcal{D}_R^2$ or $\mathcal{D}_L^2$ operators contributed by the supergraph to the corresponding term in $\Gamma[S,S^*]$.

Let us count these superderivatives. Consider a connected graph with: V_n vertices with n lines coming in; V_n^* vertices with n lines going out; I internal lines; E external incoming lines; and E^* external outgoing lines. These numbers are related by

$$I+E=\sum_n nV_n\,, \qquad I+E^*=\sum_n nV_n^*\,. \tag{30.3.2}$$

The total numbers of $\mathcal{D}_R^2$ and $\mathcal{D}_L^2$ operators are then

$$N_R=\sum_n V_n(n-1)=I+E-V=L+V^*+E-1\,, \tag{30.3.3}$$

and

$$N_L=\sum_n V_n^*(n-1)=I+E^*-V^*=L+V+E^*-1\,, \tag{30.3.4}$$

where $V=\sum_n V_n$ is the total number of vertices with incoming lines; $V^*=\sum_n V_n^*$ is the total number of vertices with outgoing lines; and $L=I-V-V^*+1$ is the number of loops. We see that a graph with any loops will have $N_R\geq E$ and $N_L\geq E^*$, so that there are at least enough $\mathcal{D}_R^2$ operators to convert all S_n into $\Phi_n=\mathcal{D}_R^2 S_n$ and its derivatives, and enough $\mathcal{D}_L^2$ operators to convert all S_n^* into $\Phi_n^*=\mathcal{D}_L^2 S_n^*$ and its derivatives. *Any graph with loops thus yields only a contribution to the four-dimensional θ-integral—in other words the D-term—of a functional of the left-chiral superfields Φ_n and their adjoints.*

The only way to obtain a contribution to an $\mathcal{F}$-term or its adjoint is to have just $N_R=E-1$ operators $\mathcal{D}_R^2$ or just $N_L=E^*-1$ operators $\mathcal{D}_L^2$, respectively. According to Eqs. (30.3.3) and (30.3.4), such a graph would have $L=0$ and $V^*=0$ or $V=0$, respectively. In other words, since we are considering only one-particle-irreducible graphs, we can only get an $\mathcal{F}$-term from a graph with a single vertex with incoming lines, and we can only get its adjoint from a graph with a single vertex with outgoing lines. This contribution is nothing but the integrated $\mathcal{F}$-term of the original superpotential, or its adjoint. Thus we see again that *there is no finite or infinite renormalization of $\mathcal{F}$-terms to any order of perturbation theory.*